

Analyzing the Impact of Pricing Strategies on Economic Fairness in Energy Communities

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Abstract—This article proposes a comprehensive framework encompassing energy sharing optimization, pricing schemes for energy transactions, and fairness metrics to validate the profit distribution of members in an energy community. First, a mixed-integer linear programming model is introduced to optimize energy sharing among the members of an energy community. Second, the study provides insights into the impact of four pricing mechanisms on economic fairness within the energy community. Finally, to validate the profit distribution, two fairness metrics assessing the quality of experience and the profit gaps are integrated to ensure economic equity within the energy community. The results shed light on the achieved level of fairness, evaluating the effectiveness of diverse pricing methods and the collaborative energy-sharing dynamics within the energy community.

Index Terms—energy community, energy sharing, fairness metrics, optimization, peer-to-peer.

I. INTRODUCTION

In recent years, a surge of interest and a clear economic motivation have driven end-consumers to unite and partake in energy sharing [1]. Particularly in the European Union, this momentum has been fueled by legislative initiatives such as the 'Clean Energy for all Europeans' package and directives on renewables and citizen energy communities [2]. These advancements mark a significant transformation, aligning with the worldwide movement towards sustainable and decentralized energy systems. The adoption of green-oriented distributed energy resources (DERs) is pivotal for a future characterized by a reduced carbon footprint by lowering CO_2 emissions, contributing to a more environmentally friendly and sustainable energy landscape [3]. Therefore, establishing the groundwork for the development of local electricity markets [4]. These markets represent a departure from traditional centralized energy systems, enabling communities to engage in sustainable energy practices, fostering collective self-consumption, and exploring innovative energy-sharing models [5]. For the successful development of these energy communities (EC), implementing a consistent and fair pricing structure that addresses environmental concerns and fosters a sense of justice among participants is essential.

Pricing mechanism transparency is fundamental for market adoption, and incorporating data protection measures is imperative. Various pricing mechanisms, such as Feed-in Tariff (FiT), mid-market rate (MMR), and supply and demand ratio (SDR), offer distinct advantages and drawbacks for governing transactions within energy communities [6]. FiT provides stability with a fixed rate but may limit flexibility, while MMR balances stability and adaptability by averaging

retailer and FiT rates. SDR introduces dynamism by resembling retailer prices until peer transactions occur, enhancing flexibility and potentially optimizing economic outcomes within the community [7]. In conjunction with pricing strategies, fairness metrics such as Quality of Experience and Minimum-Maximum fairness play a crucial role in assessing the performance of pricing mechanisms [8]. These metrics provide insights into allocation fairness, cost deviations, and energy distribution.

This article introduces an EC coupled with a pragmatic energy management framework, integrating prosumers equipped with photovoltaic (PV) generation capabilities. The study incorporates four distinct pricing mechanisms to evaluate the energy community, with a primary emphasis on mitigating fairness concerns arising from the growing prevalence of Distributed Energy Resources (DERs) within the dynamic energy landscape.

The rest of the article is organized as follows. After the introduction, Section II presents the problem formulation. Section III presents the pricing mechanisms and fairness metrics. Section IV introduces the case study. Section V presents the results and discussion. Section VI concludes the article.

II. PROBLEM FORMULATION

This section outlines the problem of optimizing surplus energy sharing within an EC. Fig. 1 illustrates the conceptual framework of the developed model.

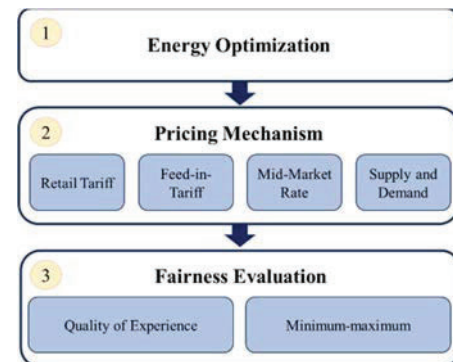


Fig. 1. Depiction of the model's proposed framework

From the optimization of energy sharing results, four pricing methods to calculate transaction costs are employed and two fairness metrics are used to assess economical fairness, aiming for an equilibrium between effective energy sharing and economic equity.

A. Market Clearing Problem

A Mixed-Integer Linear Programming (MILP) model is used to optimally determine decision variables for grid imports, exports, peer transactions, and energy bills. Additionally, binary variables are introduced to manage different conditions within the model efficiently [4],[8].

The objective function (1) is formulated to minimize energy bills, considering the energy bought from the grid, the energy sold back to the grid, and the retailer and FiT prices.

$$\min_{p_{t,h}^{\text{buyG}}, p_{t,h}^{\text{sellG}}} \left(\sum_{h \in H} \sum_{t \in T} (C_{t,h}^R \cdot p_{t,h}^{\text{buyG}} - C_{t,h}^{\text{FiT}} \cdot p_{t,h}^{\text{sellG}}) \cdot \frac{1}{\Delta t} \right) \quad (1)$$

where t represents the period, h represents the household, T the total number of periods, H the total number of households, $C_{t,h}^R$ represents the price of buying electricity from the retailer (Time-of-Use tariff), $p_{t,h}^{\text{buyG}}$ represents the amount of electricity purchased from the retailer, $C_{t,h}^{\text{FiT}}$ represents the selling price of electricity to the grid (FiT), and $p_{t,h}^{\text{sellG}}$ represents the amount of electricity sold to the retailer. The parameter Δt is employed to adapt the tariff price to the optimization time intervals, for example 15-minute increments.

The subsequent constraints cover various aspects of the energy system, including energy balance, binary conditions, and limits on energy imports and exports. Equation (2) represents the relationship between the energy bought and sold among peers in the community. It ensures that the energy bought by one peer from another equals the energy sold by that peer to the former facilitating a balanced and mutually beneficial energy exchange among the community members:

$$p_{t,h,p}^{\text{buyEC}} = p_{t,p,h}^{\text{sellEC}} \quad \forall p \neq h \quad (2)$$

where $p_{t,h,p}^{\text{buyEC}}$ is the amount household h purchased from peer p and $p_{t,p,h}^{\text{sellEC}}$ is the amount household p sold to peer h .

Equations (3-4) capture the energy exchange between community members. While (3) quantifies the total energy purchased from all peers, excluding the house itself, (4) complements this by accounting for the total energy sold to peers, also excluding any transactions involving the house under consideration. These equations provide a general view of the peer-to-peer energy transactions within the community:

$$P_{t,h}^{\text{buyEC}} = \sum_{p \neq h} p_{t,h,p}^{\text{buyEC}} \quad \forall t, h \quad (3)$$

$$P_{t,h}^{\text{sellEC}} = \sum_{p \neq h} p_{t,p,h}^{\text{sellEC}} \quad \forall t, h \quad (4)$$

where $P_{t,h}^{\text{buyEC}}$ corresponds to the total sum amount purchased in the EC market ($p_{t,h,p}^{\text{buyEC}}$), while $P_{t,h}^{\text{sellEC}}$ corresponds to the total sum amount sold in the EC market ($p_{t,p,h}^{\text{sellEC}}$).

Equation (5) ensures that the energy bought from the grid, energy bought from peers, and the energy generated from photovoltaic sources meet the demand and the energy sold to peers and the energy sold back to the grid for a specific house and time step:

$$p_{t,h}^{\text{buyG}} + p_{t,h}^{\text{buyEC}} + P_{t,h}^{\text{PV}} = D_{t,h} + P_{t,h}^{\text{sellEC}} + p_{t,h}^{\text{sellG}} \quad \forall t, h \quad (5)$$

where $P_{t,h}^{\text{PV}}$ represents the PV generation and $D_{t,h}$ corresponds

to the demand of household h .

Binary conditions play a crucial role in regulating the energy flow within the community, ensuring that certain limits are adhered to for a balanced and controlled energy exchange. The first set of binary conditions in (6-7) ensures that only one of the grid import or export and one of the peer import or export can be activated at a given time step:

$$x_{t,h}^{\text{buyGrid}} + x_{t,h}^{\text{sellGrid}} \leq 1 \quad \forall t, h \quad (6)$$

$$x_{t,h}^{\text{buyEC}} + x_{t,h}^{\text{sellEC}} \leq 1 \quad \forall t, h \quad (7)$$

where $x_{t,h}^{\text{buyG}}$ and $x_{t,h}^{\text{sellG}}$ are the binary variables for grid import and export, respectively.

Additional constraints (8-11) are used to maintain a balance in the energy transactions within the community, preventing excessive grid dependency and encouraging self-sufficiency by limiting the grid's import and export. Moreover, these restrictions ensure that peer-to-peer energy exchanges remain within predefined thresholds, fostering a fair and sustainable energy-sharing environment:

$$p_{t,h}^{\text{buyEC}} \leq p_{\text{maxbuyEC}} \cdot x_{t,h}^{\text{buyEC}} \quad (8)$$

$$p_{t,h}^{\text{sellEC}} \leq p_{\text{maxsellEC}} \cdot x_{t,h}^{\text{sellEC}} \quad (9)$$

$$p_{t,h}^{\text{buyG}} + p_{t,h}^{\text{buyEC}} \leq p_{\text{maxbuyG}} \cdot x_{t,h}^{\text{buyG}} \quad (10)$$

$$p_{t,h}^{\text{sellG}} + p_{t,h}^{\text{sellEC}} \leq p_{\text{maxsellG}} \cdot x_{t,h}^{\text{sellG}} \quad (11)$$

where p_{maxbuyEC} and $p_{\text{maxsellEC}}$ represent the maximum limit for purchasing and selling energy within the community, while p_{maxbuyG} and p_{maxsellG} correspond to the maximum permissible limit for purchasing and selling energy from and to the retailer.

B. Pricing Mechanisms

Within the EC context, the pricing of peer transactions ($C_{t,h}^{\text{EC}}$) plays a crucial role in shaping the dynamics of resource exchange. Four distinct pricing strategies were chosen to evaluate the benefits of sharing energy within the community. A Community Manager (CM) is established to implement the proposed strategies and evaluate their effectiveness in the EC.

The first strategy involves setting the peer transaction cost $C_{t,h}^{\text{EC}}$ equal to the retail energy cost, denoted as $C_{t,h}^R$. This compels EC members to obtain energy at the same rates offered by external energy retailers. While this may not offer substantial advantages to the buyer, it holds greater significance for the seller. Sellers find this strategy attractive because they receive significantly higher compensation for their energy compared to what they would have received under the FiT system.

Conversely, the second strategy explores a scenario where the peer transaction cost $C_{t,h}^{\text{EC}}$ is set equal to the FiT tariff ($C_{t,h}^{\text{FiT}}$). This stands in contrast to the first strategy, placing emphasis on favoring buyers. In this strategy, buyers enjoy the advantage of purchasing energy from peers at a lower cost compared to acquiring it from external retailers. Meanwhile, sellers receive compensation equivalent to what they would obtain from retail transactions. This approach encourages buyers to actively participate in community transactions.

In the third strategy, the peer transaction cost $C_{t,h}^{\text{EC}}$ is determined as the mid-market rate pricing ($C_{t,h}^{\text{MMR}}$), which is essentially the average between the FiT and the prevailing prices offered by external retailers, as defined in [6]. Prosumers within the EC, stand to benefit by earning

increased revenue compared to the fixed FiT. Simultaneously, they maintain a competitive advantage over standard grid consumption costs from the buyers' perspective, making energy more affordable than retailer prices. This approach enhances the appeal of peer trading for all participants, contributing to the overall benefit of the EC.

In the fourth strategy, we delve into the supply and demand ratio pricing mechanism, as outlined in [8]. This mechanism stands out as a flexible approach that adjusts to the real-time dynamics of the market, aligning local prices with the prevailing supply and demand ratios. Essentially, the supply-demand ratio pricing mechanism determines the prices in the local electricity market after the actual trades have occurred. The fundamental concept is to ensure that the price mirrors the surplus of available energy in the local market relative to the demand. This approach adds adaptability to the pricing strategy, aiming to accurately reflect the current conditions of the EC internal market.

The SDR pricing approach settles EC market prices following completed transactions, with the pricing formula (SDR_t) given in (12), which reflects the surplus available energy in the market relative to the demand.

$$SDR_t = \frac{\sum_h (P_{t,h}^{\text{sellIG}} + P_{t,h}^{\text{sellEC}})}{\sum_h (P_{t,h}^{\text{buyG}} + P_{t,h}^{\text{buyEC}})} \quad (12)$$

The standard local electricity market price, which will be used as $C_{t,h}^{\text{EC}}$, is computed using (13) as a balance between the surplus measure (s_t) and the upper and lower boundaries. These boundaries represent the buying and selling prices prevalent in the wholesale market.

$$C_{t,h}^{\text{SDR}} = (SDR_t \cdot C_{t,h}^{\text{FiT}} + (1 - SDR_t) \cdot C_{t,h}^{\text{R}}) \quad (13)$$

When surplus availability (SDR_t) is zero, indicative of a balanced market, the local price aligns with the upper bound, as defined by the sum of the selling price of surplus energy and the grid cost $C_{t,h}^{\text{R}}$. Conversely, when there is an excess surplus ($SDR_t \geq 1$) the local electricity price corresponds to the lower bound, represented by the FiT ($C_{t,h}^{\text{FiT}}$). The pricing mechanism necessitates a seamless exchange of information between market participants and the community manager, primarily focusing on net supply and demand measurements within the local electricity market.

C. Fairness Metrics

Fairness metrics are pivotal in evaluating the economic equity within local electricity markets. This becomes particularly significant in the context of market participants' diverse and occasionally irrational preferences. These metrics serve as quantitative measures to assess the extent of fairness in allocating economic benefits among community members engaging in energy transactions.

In this work, the Quality of Experience fairness (QoE) and minimum-maximum fairness (mMF) [8] were selected to evaluate the performance and fairness of the pricing mechanisms. For the computation of these metrics, the total Energy Bill (EB_h) based on the selected pricing mechanisms (C_t^{EC}) is expressed through equation (14):

$$EB_h = (C_{t,h}^{\text{R}} \cdot P_{t,h}^{\text{buyG}} - C_{t,h}^{\text{FiT}} \cdot P_{t,h}^{\text{sellIG}} + C_{t,h}^{\text{EC}} \cdot (P_{t,h}^{\text{buyEC}} - P_{t,h}^{\text{sellEC}})) \cdot \frac{1}{\Delta t} \quad (14)$$

1) Quality of Experience

The Quality of Experience (QoE) fairness metric is a quantitative measure designed to assess cost diversity among participants within the energy community. This metric provides a nuanced understanding of how pricing mechanisms impact the perceived value of energy transactions. In equation form, it is represented as (15):

$$QoE_t = 1 - \frac{\sigma_t}{\sigma_t^{\text{max}}} \quad \forall t, h \quad (15)$$

where the standard deviation of the perceived prices is denoted as σ_t reflects the extent of price variability compared to the energy bill (EB_h) divided by the difference between the energy demand ($D_{t,h}$) and the photovoltaic power ($P_{t,h}^{\text{PV}}$) (16). While the maximum deviation σ_t^{max} is provided by the subtraction of the retailer price and related taxes (C_t^{R}) by the FiT (C_t^{FiT}), in (17):

$$\sigma_t = \frac{EB_h}{D_{t,h} - P_{t,h}^{\text{PV}}} \quad \forall t \quad (16)$$

$$\sigma_t^{\text{max}} = C_t^{\text{R}} - C_t^{\text{FiT}} \quad (17)$$

A QoE_t value of 1 indicates that perceived prices are uniform across all participants, signifying a more equitable distribution of economic benefits among households. Achieving a higher QoE_t value fosters a sense of fairness and satisfaction among participants.

2) Minimum-maximum fairness

Complementarily, the Minimum-Maximum Fairness (mMF) indicator plays a pivotal role in equalizing the distribution of benefits among community participants. It is calculated as the ratio of the minimum to maximum energy bill (19):

$$mMF_t = \frac{\min_h(EB_h)}{\max_h(EB_h)} \quad \forall t \quad (19)$$

In the context of community energy sharing, a higher mMF value (approaching 1) emphasizes the minimal disparity between the participants' energy bills, showcasing an equitable sharing of economic advantages among community members.

III. CASE STUDY

The case study considers an energy community with ten active members in a suburban area enabling surplus energy sharing. Of the ten members, six are designated as prosumers, possessing PV solar panels for electricity generation.

The generation data utilized in this study was obtained from GECAD Building N in Porto, Portugal [9], analyzing one day with 15-minute intervals, totaling 96 data points. The profiles of six prosumers, derived from a common base profile, include slight random variations to simulate a range of income levels within the community. These income levels are presumed to be indicative of the ability to install additional PV generation capacity. Households 1 and 2 peak at approximately 4kW, while 3 and 4 reach around 8kW, and 5 and 6, with assumed additional PV panels, peak at nearly 12kW. Fig. 2 illustrates these diverse solar generation capacities, facilitating a comprehensive analysis of fairness implications across varied income groups.

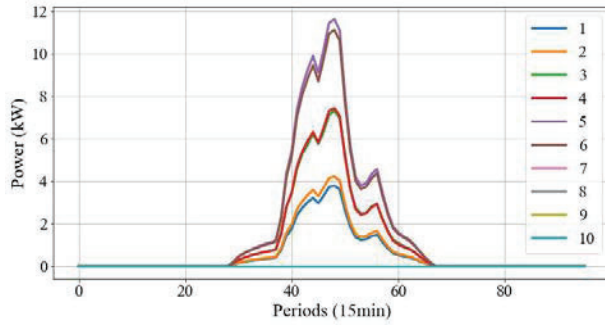


Fig. 2. Energy Community Solar Generation Profiles

The remaining households primarily function as energy consumers, with consumption profiles derived from various European households [10]. Contracted power values, following fixed powers in Portugal, allocated end-users maximum consumption levels (e.g., 3.45, 6.9, and 10.35 kW) under a three-phase contract. These diverse profiles were selected to depict the community's energy consumption dynamics and were normalized using maximum values for balanced representation.

The Business as Usual (BAU) energy bill was computed for each member as a baseline for energy bill comparisons. This calculation involved using the Demand and PV values, post self-consumption, and applying the retailer's tariff for importing energy as well as the FiT for exporting surplus energy into the grid. The BAU case works as a baseline of the costs and profits of end-users.

IV. RESULTS AND DISCUSSION

The experiments were conducted using the Python programming language within the Spyder scientific environment, part of the Anaconda Distribution. The computations were performed on a computer equipped with an Intel(R) Core(TM) i5-8250U CPU @ 1.60GHz processor, 20 GB of RAM, and running Windows 11. The Python MIP library, leveraging the Coin-or branch and cut (CBC) solver, was utilized for linear optimization during the implementation.

A. Peer transactions

Upon the initial analysis of Table I, conducted without considering pricing or fairness strategies, the post-optimization results reveal active participation in energy transactions by most members within the community. Examining specific households, Household 2 is notable for importing a significant 72.98 kW, while Household 5 actively engages in peer-to-peer imports, accounting for 25.44 kW. Household 6 emerges as the primary contributor to energy exports, showcasing its notable involvement with an impressive 80.69 kW. Further insights are gained by observing grid interactions, with Households 5 and 1 standing out for substantial grid imports of 120.67 kW and 118.89 kW, respectively. Remarkably, Household 5 also plays a key role in grid exports, contributing 64.15 kW. Finally, households 7 to 10, lacking PV generation, refrain from exporting power.

TABLE I. HOUSEHOLD POWER IMPORTS AND EXPORTS, THROUGH RETAILERS AND PEER TRANSACTIONS

Household	Import Peer ($P_{t,h}^{buyEC}$)	Export Peer ($P_{t,h}^{sellEC}$)	Import Grid ($P_{t,h}^{buyG}$)	Export Grid ($P_{t,h}^{sellG}$)
1	8.89	8.75	118.89	0.99
2	72.98	4.64	108.23	7.96
3	2.91	12.00	51.13	28.21
4	6.73	4.46	70.57	29.79
5	6.96	25.44	120.67	64.15
6	0.00	80.69	55.80	54.34
7	10.56	0.00	91.81	0.00
8	6.52	0.00	51.32	0.00
9	13.11	0.00	107.90	0.00
10	7.32	0.00	61.79	0.00

B. Pricing and bill comparison

This section provides a comprehensive examination of the results obtained from applying the four pricing methods within the EC and their impact on energy bills. Figure 3 illustrates the four pricing methods (in EUR/kWh) considered for each 15-minute interval. The C_FiT follows the FiT tariff of 0.045 EUR/kWh; the C_R correspond to the retailer triple tariff; the MMR, was derived as the average between the retailer and FiT; and finally, the C_SDR pricing profile is established using (12-13), calculated as the ratio between the energy supplied and the corresponding demand. Consequently, the C_SDR pricing plot mirrors retailer prices in the absence of peer-to-peer trading. However, when such transactions occur, the pricing decreases to levels resembling the FiT during those specific periods.

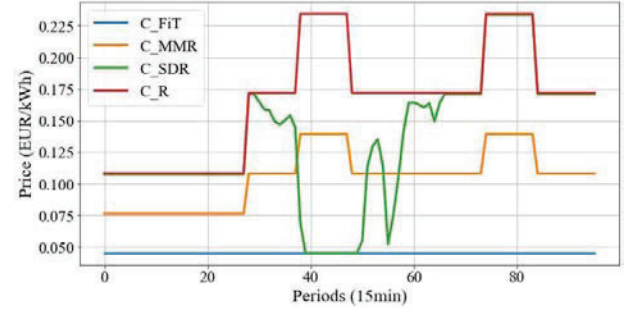


Fig. 3. Price of Peer transactions applying different pricing strategies

A comparative analysis is conducted for energy bills between the BAU scenario, and the bills calculated using the four pricing methods. The bar figure (4) portrays the energy bills for all ten households on the considered day. Notably, as PV generation rises from Low PV (1 and 2) to Mid PV (3 and 4) and High PV (5 and 6), BAU bills experience a reduction. Also, households 7 to 10, lacking PV, exhibit higher bills.

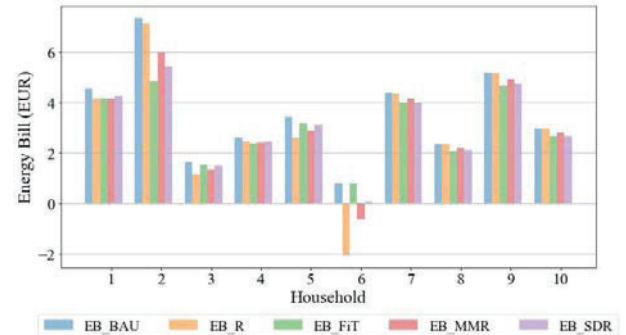


Fig. 4. Energy bill comparison using different pricing strategies

The replacement of C^{EC} with any of the four pricing methods consistently results in equal or reduced bills. The impact on Household 6 is particularly noteworthy, which significantly reduces its bill even after accounting for self-consumption and consumption during non-PV generation hours. Interestingly, Household 6, using the C^R or C^{MMR} pricing method not only achieves a negative energy bill but also receives payment for surplus energy exported to peers in the community at the same price as the retailer, making it advantageous for PV producers.

Table II presents a comparative analysis of average energy bills using distinct pricing methods across households with different PV generation levels. In the Low PV scenario, C^{FIT} achieves the most significant reduction (24.37%). For Mid-PV households, all methods contribute to reduced bills, with C^R leading at 15.09%. In High PV scenarios, C^R remarkably lowers bills by 86.32%, outperforming other methods. Households without PV production similarly experience either consistent or reduced energy bills when all the pricing strategies are implemented.

TABLE II. AVERAGE ENERGY BILL COMPARISON

	Low PV	Mid-PV	High PV	No PV
BAU	5.95	2.12	2.12	3.72
$C_{t,h}^R$	5.66	1.8	0.29	3.72
	-4.87%	-15.09%	-86.32%	0%
$C_{t,h}^{FIT}$	4.5	1.97	2	3.35
	-24.37%	-7.08%	-5.66%	-9.95%
$C_{t,h}^{MMR}$	5.08	1.89	1.15	3.53
	-14.62%	-10.85%	-45.75%	-5.11%
$C_{t,h}^{SDR}$	4.85	1.98	1.6	3.37
	-18.49%	-6.60%	-24.53%	-9.41%

C. Fairness Metric

Figures 5 and 6 illustrate the fairness metrics, specifically QoE and mMF, as they are applied to energy bills using the four pricing methods.

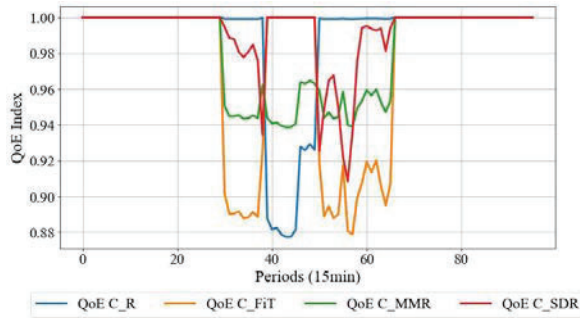


Fig. 5. QoE indicator for different pricing strategies

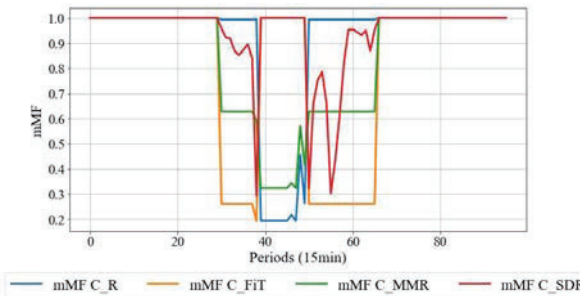


Fig. 6. mMF indicator for different pricing strategies

During non-photovoltaic generation periods, fairness values hover around 1, signifying equitable conditions as all participants engage with the retailer. However, during trading periods, disparities emerge, with C^{SDR} exhibiting greater fairness compared to C^{FIT} . Notably, C^{SDR} stands out as a more equitable choice, approaching a fairness metric value of 1 in both QoE and mMF. This characteristic underlines the potential of C^{SDR} to achieve fairness for both PV sellers and peer buyers, making it a favorable option.

V. CONCLUSION

In conclusion, this study extends beyond mere cost optimization by incorporating diverse price mechanisms and fairness metrics, emphasizing the critical role of fair energy cost distribution within energy communities. Balancing economic efficiency and fairness is essential for building trust and ensuring long-term sustainability in these communities. The findings highlight the importance of considering such metrics when designing pricing mechanisms for decentralized energy systems, offering insights into creating more equitable and user-friendly local energy markets. Future work will explore larger communities, scalability, dynamic pricing methods, and additional fairness metrics to further enhance the effectiveness of pricing strategies.

REFERENCES

- [1] J. Li, Y. Ye, D. Papadaskalopoulos, and G. Strbac, 'Computationally Efficient Pricing and Benefit Distribution Mechanisms for Incentivizing Stable Peer-to-Peer Energy Trading', *IEEE Internet Things J.*, vol. 8, no. 2, pp. 734–749, Jan. 2021, doi: 10.1109/JIOT.2020.3007196.
- [2] J. Lowitzsch, C. E. Hoicka, and F. J. van Tulder, 'Renewable energy communities under the 2019 European Clean Energy Package – Governance model for the energy clusters of the future?', *Renew. Sustain. Energy Rev.*, vol. 122, p. 109489, Apr. 2020, doi: 10.1016/j.rser.2019.109489.
- [3] T. Dias de Lima, J. Soares, F. Lezama, J. Franco, and Z. Vale, 'A Risk-Based Planning Approach for Sustainable Distribution Systems Considering EV Charging Stations and Carbon Taxes', *IEEE Trans. Sustain. Energy*, vol. PP, pp. 1–14, Oct. 2023, doi: 10.1109/TSTE.2023.3261599.
- [4] R. Faia, F. Lezama, J. Soares, T. Pinto, and Z. Vale, 'Local electricity markets: A review on benefits, barriers, current trends and future perspectives', *Renew. Sustain. Energy Rev.*, vol. 190, p. 114006, Feb. 2024, doi: 10.1016/j.rser.2023.114006.
- [5] M. Gržanić, T. Capuder, N. Zhang, and W. Huang, 'Prosumers as active market participants: A systematic review of evolution of opportunities, models and challenges', *Renew. Sustain. Energy Rev.*, vol. 154, p. 111859, Feb. 2022, doi: 10.1016/j.rser.2021.111859.
- [6] V. Z. Gjorgievski, S. Cundeva, N. Markovska, and G. E. Georghiou, 'Virtual net-billing: A fair energy sharing method for collective self-consumption', *Energy*, vol. 254, p. 124246, Sep. 2022, doi: 10.1016/j.energy.2022.124246.
- [7] M. B. Roberts, A. Sharma, and I. MacGill, 'Efficient, effective and fair allocation of costs and benefits in residential energy communities deploying shared photovoltaics', *Appl. Energy*, vol. 305, p. 117935, Jan. 2022, doi: 10.1016/j.apenergy.2021.117935.
- [8] M. F. Dyrge, K. Berg, S. Bjarghov, and Ü. Cali, 'Local electricity market pricing mechanisms' impact on welfare distribution, privacy and transparency', *Appl. Energy*, vol. 341, p. 121112, Jul. 2023, doi: 10.1016/j.apenergy.2023.121112.
- [9] C. Goncalves, R. Barreto, P. Faria, L. Gomes, and Z. Vale, 'Dataset of an energy community's generation and consumption with appliance allocation', *Data Brief*, vol. 45, p. 108590, Dec. 2022, doi: 10.1016/j.dib.2022.108590.
- [10] R. Faia, C. Goncalves, L. Gomes, and Z. Vale, 'Dataset of an energy community with prosumer consumption, photovoltaic generation, battery storage, and electric vehicles', *Data Brief*, vol. 48, p. 109218, Jun. 2023, doi: 10.1016/j.dib.2023.109218.